Printed Pages– 4

Roll No. 3633037 1902

D037731(037)

B. Tech. (Seventh Semester) Examination, Nov.-Dec. 2022

(New Scheme)

(Mechanical Engg. Branch)

MACHINE TOOL TECHNOLOGY

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : *Attempt all questions. Part (a) of each question is compulsory and carries 4 marks each. Attempt any two parts from (b), (c) and (d) and having 8 marks each.*

Unit-I

1. (a) Shwo the geometry of singl point cutting tool by ASA and ORS.

(b) Explain mechanism of metal cutting. Differentiate

between orthogonal and oblique cutting by giving suitable examples and neat sketches.

(c) What is chip formation and how many types of chip formed during metal cutting? Explain them.

(d) With the help of neat sketch describe the different angles in metal cutting.

Unit-II

2. (a) What is mechinability? Define machinability index.

(b) What are the sources of heat in metal cutting? Also explain different types of cutting fluid and function of cutting fluid.

(c) Explain the mechanism of tool wear. Differentiate between creter wear and flank wear.

(d) Discuss Taylor's relationship for cutting speed tool life.

Unit-III

3. (a) What are the essential requirements of machine tool.

(b) Analyse the various forces for the headstock, tailstock and saddle of a lathe.

(c) Explain various types of guidways used in a machine tool.

(d) Write down down the design procedure and advantages of antifriction guidways.

Unit-IV

4. (a) Explain :

(i) Ray diagram

(ii) Speed Gear Box

(b) Design gear box to privide output speed ranging from ISO rpm and 800 rpm. Where input speed is 600 rpm.

(c) What is speed year box? Explain the function and working of speed gear box.

(d) Explan the procedure for selecting speed step in stepped drive.

Unit-V

5. (a) What is feed gearbox. Write the name the elements of feed gear boxes.

(b) Discuss the following :

(i) Feed gearbox of Meander's type

(ii) Norton gearbox

(c) Describe machine tool maintenance.

(d) Briefly explain the acceptance test carried out on lathe machine.